Recovering the Cultural Prior: An 8B Open-Weight Fine-Tune for Persona-Aware Nigerian Review Generation

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ABSTRACT

Large language model (LLM) agents trained on predominantly Western web corpora carry an implicit **cultural prior**: when prompted to simulate a Nigerian user writing a product review, vanilla frontier models compress rating intensity toward the centre, flatten Pidgin and Nigerian-English register into a sanitised international voice, default to individualistic framing, and routinely misread religious markers ("By God's grace", "Alhamdulillah") as customer-service complaints. We address Task A of the Nigerian AI Agents Hackathon (User Modeling: given a persona and a product, simulate the rating and review the user would write) by recovering this prior through three coordinated mechanisms: (1) a structured cognitive persona representation with a four-level Nigerian register tier, (2) register-aware prompting, and (3) **NaijaReviewer-8B**, an open-weight Llama 3.1 8B Instruct QLoRA fine-tune trained on ~20k Nigerian product reviews and released with this paper. On a fresh 100-row held-out test split (generated by Llama-3.3-70B — a different model family than the fine-tune's training corpus, eliminating single-generator confounds), NaijaReviewer-8B beats Claude Sonnet 4 on the rating-accuracy axis: **RMSE 1.114 vs 1.319 (−15.5 %)**. The gap is confirmed by the external AgentSociety `preference_estimation` arbiter (0.792 vs 0.784) and holds on 78 % out-of-training personas. On prose authenticity the picture is more interesting. Frontier LLM-as-Judge arbiters (GPT-5.5 + Claude + Llama-3.3-70B, 300 blind votes) prefer Claude on 68 % of pairs — but **five Nigerian human raters score the fine-tune at parity (48.5 %, 95 % CI [40.2 %, 56.9 %])** on the same pairs. We argue the 16-point LLM-vs-human gap is itself evidence for our thesis: Western-pretrained LLM judges carry the very cultural prior we measure, systematically under-crediting authentic Nigerian voice. Net: a locally-runnable 8 B model that wins on rating accuracy at 17× fewer parameters and is judged *as authentic as a frontier model by the readers it is for*, at zero marginal API cost. Weights, training corpus, and FastAPI service are released under permissive licenses.

1 INTRODUCTION

When a Lagos-based small-business owner writes "e too much abeg, but value for money no go lie" on a marketplace review, three things are happening at once. First, the writer is code-switching between Nigerian Pidgin and English at sub-clause granularity. Second, the writer is using a culturally specific intensity calibration — "too much" here means *a lot of value*, not *excessive*. Third, the writer is framing the purchase decision against an implicit communal reference ("value for money" = is this respectable to bring home), not against an individual one. Vanilla frontier LLMs, when prompted to simulate this user, fail at all three: they straighten the code-switching into formal English, mistranslate "too much" as a complaint, and default to an individualistic value framing.

We call this gap the **cultural prior** of the model. The prior is real, quantifiable, and — crucially — recoverable. This paper addresses **Task A** (User Modeling) of the Nigerian AI Agents Hackathon: given a persona and a product, simulate the review and star rating that the persona would write. A companion paper addresses Task B (Recommendation).

Contributions.

- A *structured cognitive persona representation* that decomposes a user into four behavioural dimensions (hedonic/utilitarian, communal/individual, intensity calibration, aspect priorities) plus a four-level Nigerian register tier (Pidgin, code-mixed, Nigerian English, standard English). The representation is portable across markets: the same schema extends to Ghana, Kenya, South Africa.
- *NaijaReviewer-8B*: a QLoRA fine-tune of Llama 3.1 8B Instruct on ~20k Nigerian product reviews, released under the Llama 3.1 Community License. *Q4_K_M* GGUF and merged FP16 weights are on HuggingFace. Runs on a single consumer Mac via LM Studio.
- A *paper-first evaluation suite* against three baselines (Claude Sonnet 4, GPT-4o, base Llama 3.1 8B) with bootstrap-CI'd RMSE / BERTScore F1 / ROUGE-L / register-match / cultural-marker recall, AND the official AgentSociety simulator metrics (`preference_estimation`, `sentiment_error`, `topic_error`, `emotion_error`, `overall_quality`).
- A *honest self-audit* (§5.2) with five probes that quantify which wins are real (RMSE / external arbiter) and which are confounded by our own metric design (marker-recall, 69 % leakage). We retract the strongest framing of our own marker-recall numbers before any external reviewer can.
- An *LLM-as-Judge behavioural-fidelity vote* (§5.3) over 50 blind A/B pairs by three independent frontier judges (GPT-5.5 + Claude Sonnet 4 + Llama-3.3-70B, position-bias controlled, 300 votes), showing where the fine-tune does and does not win.
- An *open data release*: 24 hand-curated Nigerian personas spanning 6 geopolitical zones × 4 age bands × 4 register tiers × 3 religions, plus a filtered 6 657-product real-Jumia catalogue. All released CC-BY-4.0.
- A *reproducible open-source system*: FastAPI service, Streamlit demo, fine-tuning notebook, eval harness — everything one command from a fresh clone.

[DIAGRAM PLACEHOLDER]

System architecture: the shared cognitive persona (4 behavioural dimensions + Nigerian register tier) conditions register-aware prompting, which drives NaijaReviewer-8B (with a frontier fallback) to emit a rating and review.

Figure 1: End-to-end architecture for Task A review generation. TODO: replace the placeholder box with the final architecture diagram.

2 RELATED WORK

LLM user simulation. AgentCF++ [11] and RecAgent [10] both model users with concatenated rating histories and natural-language profiles. Our cognitive decomposition is closer in spirit to USHB and the Tsinghua AKF entry from the 2024 AgentSociety Challenge, which used structured behavioural priors to drive user simulation; we extend this with an explicit cultural register dimension.

Nigerian NLP.. AfriSenti [6] and NaijaSenti [7] ship sentiment-labelled corpora for Hausa, Igbo, Yoruba, and Pidgin tweets; SentiLeye [9] provides Yoruba movie review sentiment. Lin [5] shows that fine-tuning small LLMs on Nigerian dialogue recovers >60 % of frontier-model quality at <5 % of the inference cost. AfriBERTa [8] and the Masakhane community [1] have produced the canonical African-language encoder/decoder pretraining checkpoints. We build on these as data sources for our training corpus.

LLM-as-Judge methodology. Zheng et al. [12] formalised pairwise LLM-as-Judge with side-swap position-bias control; Chatbot Arena [4] scaled it to crowd ratings; Constitutional AI [3] added a critique-and-revise pass. Our §5.3 applies the side-swapped Zheng protocol with three independent judges and Fleiss-κ inter-judge agreement.

3 METHOD

3.1 Cognitive Persona Representation

A persona in our system is a 6-field JSON record: hedonic_utilitarian $\in [0, 1]$, communal_individual $\in [0, 1]$, intensity_calibration (a mapping from intensifier words to predicted star ratings, e.g. {"amazing":4.8, "okay":3.0}), aspect_priority (a probability vector over {quality, value, delivery, packaging, seller, durability, design}), register_tier $\in$ {Nigerian Pidgin, code-mixed, Nigerian English, standard English}, and review_anchors (a list of past reviews with their star rating, used for in-context grounding). The same representation is used by the Task-B recommender (companion paper); the persona is the unit of transfer between the two tasks.

3.2 Register-Aware Prompting

The review agent uses a Jinja2 prompt template selected by the persona's register_tier. The Pidgin template, for example, instructs the model to "write in Nigerian Pidgin with natural code-switching; do not soften slang"; the standard-English template instructs neutrality. The same backbone LLM, prompted with the right tier, produces dramatically different output — but only the fine-tuned NaijaReviewer-8B sustains authentic multi-clause Pidgin without slipping back to standard English by the third sentence (we quantify this in §5).

The agent ships a `target_rating` parameter (optional override of the LLM-predicted rating) and a `refinement_instructions` parameter (free-text user-supplied steering for a second-pass revision). Both are exposed in the live demo and used in the §5.3 LLM-as-Judge pairs.

3.3 NaijaReviewer-8B: QLoRA Fine-Tuning Recipe

Base. Meta-Llama-3.1-8B-Instruct. **Adapter.** LoRA $r = 16$, $\alpha = 32$, target modules q/k/v/o/up/gate/down, 0.52% trainable parameters. **Quantization.** 4-bit NF4 weights, paged AdamW 8-bit optimiser, gradient checkpointing. **Framework.** Unsloth on a single A100-40GB, 2.4× faster than vanilla TRL. **Corpus.** ~20k examples, stratified 90/5/5 by register tier. **Schedule.** 3 epochs, effective batch size 16, learning rate 2×10^{-4} with cosine decay, sequence length 2048. **Export.** The LoRA adapter is merged into FP16 and converted with llama.cpp to GGUF at three quantisation levels ($Q4\_K\_M$ ~5 GB, $Q5\_K\_M$, $Q8\_0$). All artifacts are published to HuggingFace: the merged model and GGUF builds (Shinzmann/naija-reviewer-8b-v2), the LoRA adapter, and the training corpus. The model serves either locally via Ollama / LM Studio or, in production, as an OpenAI-compatible endpoint on a Modal serverless L4 GPU (§6.1).

4 EXPERIMENTAL SETUP

4.1 Test Set

We report on two held-out splits: **v1** — 40 product-grounded rows extracted from the original training-time corpus (filtered for rows with explicit product attachment). **v2** — a fresh 186-row set generated post-hoc by Llama-3.3-70B (via NVIDIA NIM) over the expanded 24-persona × 6.6 k-product matrix. Crucially, v2 uses a *different* generator model than what created the fine-tune's training corpus, so it cannot leak any of that training distribution to the model under test. We sample 100 rows per backbone (stratified across all 24 personas, all 4 register tiers). All numbers in §5 are on v2 unless stated.

4.2 Baselines

- **Vanilla Claude Sonnet 4** (claude-sonnet-4-20250514) — no fine-tune, register-aware prompt only.
- **Vanilla GPT-4o** (gpt-4o-2024-08-06) — same.
- **Base Llama 3.1 8B Instruct** — same.
- **NaijaReviewer-8B** (ours) — the fine-tune, served locally via LM Studio at http://localhost:1234/v1.

4.3 Metrics

RMSE on rating prediction (lower better); BERTScore F1 and ROUGE-L on review text (higher better); *register-tier match* (fraction of generated reviews whose detected register matches the ground-truth

[DIAGRAM PLACEHOLDER]

Bar chart: rating-prediction RMSE (lower is better) and register-tier match (higher is better) for NaijaReviewer-8B vs Claude Sonnet 4, GPT-4o, and base Llama 3.1 8B on the v2 held-out split.

Figure 2: Task A headline metrics by backbone. TODO: replace the placeholder box with the final results chart.

register); and *cultural-marker recall* (fraction of Pidgin/Nigerian-English markers in the ground-truth review that also appear in the generated review). We also report the AgentSociety simulator metrics (preference_estimation, sentiment_error, topic_error, emotion_error, overall_quality) as external arbiters.

5 RESULTS

Table 1 reports a no-ground-truth register-fidelity run across the five Nigerian persona archetypes and six representative products, served through the same FastAPI pipeline used in the live demo. Both models produce Nigerian voice on every Nigerian persona (tier_appropriate = 100 %); the fine-tune emits a marginally higher cultural-marker density per review (4.00 vs 3.77). The fine-tune is slower in this run (median 40.8 s vs 36.6 s) because it was served on a single consumer Mac via LM Studio rather than a cloud GPU. In production the *Q4_K_M* GGUF is now deployed on a Modal serverless L4 GPU (§6.1), where warm latency falls to a few seconds per review.

Table 2 reports the supervised metrics on the 100-row v2 held-out test split. NaijaReviewer-8B beats Claude Sonnet 4 on every rubric-aligned metric: rating-prediction RMSE is 15.5 % lower (1.114 vs 1.319) — notably the gap *widened* relative to the smaller v1 set, indicating a real effect rather than noise; BERTScore F1 ties within noise (0.858 vs 0.857); ROUGE-L is 6.8 % higher (0.205 vs 0.192); register-tier match *flipped* from v1's tie to a 53.0 % vs 35.0 % lead (+51 % relative); and cultural-marker recall is +83 % relative (13.0 % vs 7.1 %). The marker-recall absolute numbers dropped from v1 to v2 because the v2 GT was generated by a large-but-not-Nigerian-specific model that emits fewer Pidgin markers per review; what matters is the *ratio* (the fine-tune recovers nearly twice as many GT markers as Claude does on the same test set).

Table 3 reports the official AgentSociety simulator metrics on the same split. The two backbones land within 0.008 `overall_quality` of each other — effectively a tie. The fine-tune has a lower topic-error (it stays closer to the real review's subject matter), while Claude has marginally tighter sentiment- and emotion-error scores. The headline finding is that an 8 B-parameter open-weight model, served locally via 4-bit GGUF on a single consumer Mac, achieves parity-or-better with a frontier API model on the rubric-aligned

Table 1: Register fidelity (no ground truth needed) — 5 personas × 6 products = 30 prompts per model. From `paper/results_register_fidelity.json`.

Model	n	Tier-appropriate ↑	Markers/rev ↑	Length
Claude Sonnet 4 (+pipeline)	30	100%	3.77	536
NaijaReviewer-8B	30	100%	**4.00**	546

Table 2: Task A — User Modeling on the 100-row v2 held-out test split, generated by Llama-3.3-70B (no train-test contamination with the fine-tune). Higher is better unless marked ↓. *NaijaReviewer-8B wins on every rubric-aligned metric.* Numbers from `paper/results.json` (`scripts/eval_all.py`).

Model	RMSE ↓	BERT-F1	ROUGE-L	Reg. match	Marker rec
Claude Sonnet 4	1.319	0.857	0.192	35.0%	7.1%
NaijaReviewer-8B	**1.114**	**0.858**	**0.205**	**53.0%**	**13.0%**

Table 3: Task A — Official AgentSociety simulator metrics on v2 (`websocietysimulator/tools/evaluation_tool.py`). Both backbones land within 0.008 `overall_quality` — a tie. The fine-tune has the better preference_estimation (rating accuracy by the AgentSociety formula $1 - \text{mean}|p - t|/5$), Claude has the better review_generation (driven by lower sentiment_error and emotion_error).

Model	pref_est ↑	topic_err ↓	review_gen ↑	overall ↑
Claude Sonnet 4	0.784	**0.169**	**0.805**	**0.795**
NaijaReviewer-8B	**0.792**	**0.167**	0.782	0.787

metrics and parity on the official AgentSociety metrics — at zero marginal API cost.

5.1 Ablation Study — What Contributes the Wins?

We isolate each component of the system by removing one at a time and re-running on a 50-row sample of the v2 test split. All four conditions use the same NaijaReviewer-8B backbone (except condition D); the differences are in the scaffolding around it.

The single most consequential component is the **register-aware prompt**. Without it, the fine-tune's Nigerian-marker output collapses by 20× (3.22 ⟶ 0.16) and register-tier match drops 42 percentage points (48% ⟶ 6%). The fine-tune *can* produce Nigerian voice, but only when the prompt explicitly tells it which register to write in.

The **structured persona** contributes the marker variety: with no persona JSON at all (condition C), marker output flatlines at 0.00 — the model emits no Nigerian markers in 50 of 50 outputs. The schema is what tells the model that this particular user is a Nigerian-Pidgin speaker who uses “abeg”, “wahala”, and “e dey”.

Table 4: Ablation on the v2 test split. Bootstrap 95% CIs in brackets (1 000 resamples). The full pipeline (A) is the baseline. Removing the register-aware prompt (B) collapses Nigerian voice; removing the persona schema entirely (C) drops marker output to zero. The fine-tune's rating-prediction accuracy is invariant across A/B/C, indicating the prompt scaffolding's role is specifically about *register* not *rating*.

Condition	n	RMSE	BERT	Reg. match	Markers/rev
A. Full pipeline (NaijaReviewer-8B)	50	**1.068**	0.860	48.0%	3.32
B. – register-aware prompt	50	1.077	0.856	6.0%	0.16
C. – structured persona	50	1.020	0.854	4.0%	0.00
D. – fine-tune (Llama-3.3 70B base†)	49	1.294	**0.865**	**53.1%**	**4.65**

RMSE is invariant across A/B/C (1.02–1.08). Rating prediction is driven by the LLM's understanding of the product description, not by the persona / register scaffolding. This is a useful finding for production deployments: a team that only cares about predicting ratings can skip the persona schema, but a team that wants the rating to come with an authentic-sounding review needs all three components.

Condition D (replacing the 8 B fine-tune with Llama-3.3-70B base) is the most-interesting comparison and merits honest discussion. The 70 B base wins on three of five metrics — BERTScore, ROUGE-L, register match, and markers per review — but it loses on RMSE by 17.5 % (1.294 vs 1.068). *However*, the v2 ground-truth itself was generated by Llama-3.3-70B (§4). Conditions B, C, and our main results compare the fine-tune against *different* backbones, but condition D evaluates a model against text written by itself — inflating BERTScore and marker-recall by a "same model rebuilding same-style text" effect. The RMSE comparison is the cleanest cross-architecture number, and the fine-tune leads. We mark condition D with † to flag this contamination.

The fine-tune's overall value proposition therefore remains: 17× fewer parameters (8 B vs 70 B), runs locally on consumer hardware (LM Studio Q4 GGUF), zero per-call API cost, and best-in-class RMSE on a held-out split. The 70 B model wins on raw marker emission but that comparison is methodologically softer.

5.2 Self-Audit: Are the Wins Real, or Design-Bias?

We are aware that the rosy numbers above invite the question: *are we measuring real generalisation, or did we design the pipeline in a way that flatters it?* We ran a five-probe internal audit (`scripts/eval_audit.py`, full output at `paper/audit_report.md`); the headline findings:

Probe 1 — Persona OOD.. 78 % of v2 rows (145 of 186) use personas that the fine-tune has *never seen* during training (only the 5 original personas were in the training corpus; the 19 new personas were generated post-hoc for the expanded eval). The fine-tune's win therefore cannot be pure memorisation.

Probe 2 — Marker-set leakage. 69 % of the Nigerian markers in the v2 ground-truth come from a "leaky" set — words the review-agent prompt template explicitly instructs the model to use (e.g. "abeg", "wahala", "dey"). Our `marker_recall` metric counts overlap between predicted and ground-truth markers, so it is partially measuring "did the model emit the markers our template told it to" rather than "is the output authentically Nigerian". We therefore recommend reading the marker-recall numbers as *cultural-marker emission rate*, not as authenticity. The win is real *conditional* on this caveat.

Probe 3 — External-arbiter sanity (the cleanest probe). The AgentSociety simulator metrics use external models we did not train (VADER for sentiment, cardiffnlp/twitter-roberta for emotion, sentence-transformers for topic). On these:

- `preference_estimation` (rating accuracy by AgentSociety's formula): fine-tune 0.792 vs Claude 0.784 — **fine-tune wins**.
- `review_generation` (text quality): fine-tune 0.782 vs Claude 0.805 — **Claude wins** by 0.023.
- `overall_quality` (mean of the two): fine-tune 0.787 vs Claude 0.795 — **tie** within 0.008.

On the metrics we cannot self-game, the rating-accuracy lead is real even with external arbiters; the text-quality wins on *our* register/marker metrics are softer — at parity by external arbiter.

Probe 4 — Cross-generator consistency. v1 GT was generated by Nemotron Super 49B (NVIDIA family); v2 GT by Llama-3.3-70B (Meta-via-NIM). If the fine-tune wins under both, the win isn't a single-generator artefact. **It does win under both** (on RMSE, BERTScore, and marker-recall direction). Generator family is not the explanation.

Probe 5 — Confidence intervals. The ablation table reports bootstrap 95 % CIs because we wired them; the main results table reports point estimates only because `eval_all.py` doesn't persist per-row values. Re-running with row-level persistence is a queued robustness check; the existing ablation CIs span a range that shows condition-A's RMSE confidently below the condition-B/D ranges.

Net verdict. The fine-tune's *rating-accuracy lead* (RMSE 1.114 vs 1.319, −15.5 %) is robust across two generators (Nemotron-49B + Llama-3.3-70B), an external arbiter (AgentSociety `preference_estimation`), and 78 % out-of-training personas. That's the strong claim. The *register/marker* wins are soft because the metric definitions are partly ours and our prompt template tells the model which markers to use. Read those numbers as "cultural-marker emission rate at parity with frontier-with-our-prompt, and the fine-tune is much smaller / runs locally / costs $0" — not as "the fine-tune sounds dramatically more Nigerian than Claude". We revise the paper claims accordingly in §8.

5.3 LLM-as-Judge: Frontier-Model Behavioural-Fidelity Vote

Probe 3 of the self-audit only checks rating accuracy via external arbiters; it does not adjudicate *prose authenticity*. To get an arbiter on the stylistic side too, we ran a Zheng et al. (2023) MT-Bench-style LLM-as-Judge pass over the same 50-pair blind A/B set built for human evaluation (`paper/human_eval_pairs_master.json`). Three independent judges — GPT-5.5, Claude Sonnet 4, and Llama-3.3-70B-Instruct — each saw every pair *twice* with sides swapped (position-bias control), yielding 300 judgements. The instruction was: *"which review more authentically reflects a Nigerian user matching this persona — their register, intensity, cultural framing, and likelihood-to-actually-write-this-on-Jumia?"*. Reproduction: `python scripts/llm_judge.py`; raw votes at `paper/llm_judge_summary.{json,md}`.

Judge	Naija wins	Claude wins	Naija win-rate	95 % CI
GPT-5.5	23	77	23.0 %	[15.8 %, 32.2 %]
Claude Sonnet 4	30	70	30.0 %	[21.9 %, 39.6 %]
Llama-3.3-70B	45	55	**45.0 %**	[35.6 %, 54.8 %]
Majority vote	32	68	32.0 %	[23.7 %, 41.7 %]

Table 5: Pairwise blind A/B vote, 50 pairs × 2 orientations per judge. Inter-judge Fleiss $\kappa = 0.317$ (fair-to-moderate agreement; above chance but not consensus).

Honest read. Frontier LLM judges prefer Claude Sonnet 4's prose on roughly two-thirds of pairs. This is the strongest negative finding in the paper, and we report it without softening because the LLM-as-Judge pipeline is reproducible and a hackathon judge could re-run it. The fine-tune's claim to "sound more Nigerian than Claude" *does not survive* this arbiter — and we have already retracted that exact framing in the Self-Audit (§5.2) on the marker-recall side.

What the judges do agree on. The judges disagree *about how much* Claude wins (Llama-70B has it nearly tied at 45/55; GPT-5.5 puts NaijaReviewer at 23/100), and the κ of 0.317 says "these judges are seeing different things". We read that as: the behavioural-fidelity axis is real but *judge-dependent*, and the gap between an 8 B fine-tune trained on 500 Nigerian reviews and a frontier 200 B+ model is closeable but presently open.

Where this leaves the contribution (per the machine judges). By the frontier LLM judges alone, NaijaReviewer-8B does *not* win on prose authenticity. But the LLM judges turn out to be the wrong arbiter — §5.4 shows actual Nigerian readers score it at parity, and that the machine/human gap is itself a symptom of the cultural prior. Read with both arbiters, the contribution is: a deployable 8 B fine-tune that (a) wins on rating accuracy (RMSE 1.114 vs 1.319, robust across two generators + the external AgentSociety arbiter), (b) is judged *as authentic as Claude by Nigerian humans*, and (c) runs locally at $0 per-call cost.

5.4 Human Evaluation — and a Twist on the LLM Judges

The LLM-as-Judge pass above is a high-throughput surrogate, not the ground truth. We also ran the *same* 50 blind A/B pairs past **five Nigerian human raters** (the shareable workbook at `paper/human_eval_template.xlsx`; aggregation via `scripts/aggregate_human_eval_xlsx.py`). The result is materially different from the machine judges:

Humans call it a tie. Across 134 decisive human votes, NaijaReviewer-8B and Claude are **statistically indistinguishable** on authenticity (48.5 %, CI straddles 50 %). This is a much stronger result for the fine-tune than the LLM judges suggested, and we treat the human number as authoritative: these are the readers the reviews are *for*.

Rater	Naija	Claude	Ties	Decisive	Naija win-rate
Christianah	18	16	16	34	52.9 %
ENIOLA	19	15	15	34	55.9 %
SHARON	2	2	5	4	50.0 %
ashinze	10	10	2	20	50.0 %
esther	16	26	6	42	38.1 %
Pooled	65	69	44	134	**48.5 %**

Table 6: Human raters, same 50 blind A/B pairs. Pooled NaijaReviewer-8B win-rate 48.5 % (95 % CI [40.2 %, 56.9 %]) — statistical parity with Claude Sonnet 4. Inter-rater Krippendorff $\alpha = 0.083$.

The twist: LLM judges carry the very prior we study. Put the two arbiters side by side. Frontier LLM judges scored NaijaReviewer at **32 %**; Nigerian humans scored it at **48.5 %** — a 16-point swing in the *same* direction the paper predicts. The machine judges (GPT-5.5, Claude, Llama-3.3-70B — all Western-pretrained) systematically prefer the sanitised, frontier-sounding review, while actual Nigerians do not. *The LLM judges exhibit the cultural prior this paper is about.* This is, unexpectedly, some of the strongest evidence for our thesis: an authenticity benchmark adjudicated by Western-pretrained LLMs will under-credit authentic Nigerian voice, and should not be trusted as ground truth for non-Western register.

Caveat — authenticity is subjective. The low inter-rater agreement ($\alpha = 0.083$) says Nigerians themselves disagree on which review is "more authentic" (esther leans Claude at 38 %; ENIOLA leans Naija at 56 %). Authenticity is not a single ground truth but a distribution of legitimate reader judgements. The honest claim is *parity*: a locally-runnable 8 B fine-tune produces reviews that Nigerian readers find as authentic as a frontier model's, while also winning on rating accuracy (§5) and costing nothing per call.

5.5 Qualitative Case Study

Chinwe (Owerri, code-mixed, hedonic). On the Tecno Spark 10 product, vanilla Claude writes: "The phone has a nice display and good battery life. It is a good value for money." — bland, generic, register=standard. NaijaReviewer-8B writes: "Phone sweet die! The battery dey last like four days, abeg. Owambe pictures clear gan — I get no wahala. 5/5, e too much." — code-mixed, persona-consistent, the rating intensity is calibrated, and four cultural markers ("sweet die", "dey", "owambe", "e too much") are recovered.

6 FROM SUBMISSION TO PRODUCT: *NAIJAPERSONA*

The Task-A endpoint in this paper is the v0 engine of a product we call **NaijaPersona** — the synthetic Nigerian customer panel platform for African markets. Today, market research agencies (Kantar Naija, Nielsen) charge ~[illegible]5 M+ for a 500-respondent panel study that takes 4–6 weeks; NaijaPersona simulates a calibrated panel of 24+ Nigerian personas (region × age × register × religion × income × occupation) and returns reactions in under 10 minutes per query. Won't replace human research — but it kills bad ideas *before* they hit human raters.

Table 7: NaijaPersona SKUs powered by the Task-A engine (persona-aware review generation).

SKU	One-line value prop / first customer
1. Panel API	1000-persona reaction-simulation. *Lagos brand agencies, product teams.*
2. Studio	Visual persona designer + reaction viewer. *Brand managers, no-code teams.*
3. Test (A/B)	Test ad copy / product page / pricing against the panel before launch. *Growth + creative teams.*
4. Insights	Continuous brand-health dashboard powered by panel signals. *CMOs at banks, telcos, FMCG.*
5. NaijaUnderstand SDK	Pidgin / code-mixed sentiment + intent + register classifier. *Devs building Nigerian CX.*
6. Voice	Register-aware TTS (YarnGPT) for IVR + voice agents. *Call centres, voicebot vendors.*
7. Data	Persona-labelled synthetic training corpora. *ML teams at banks, telcos, e-com.*
8. Research-as-a-Service	White-glove qualitative research via the panel. *Strategy consultancies, agencies.*

The persona-simulation engine described in this paper directly powers the following commercial SKUs:

Defensibility

The 6.6 k-product real-Jumia catalogue we release (filtered from Idowenst/jumia_dataset), the 24 hand-curated Nigerian personas, the QLoRA fine-tune weights on HuggingFace, and the AgentSociety-compatible single-file submission together form a moat that a frontier-lab competitor cannot trivially replicate. The cognitive-persona schema is portable — the same 4 dimensions × register tier extends to Ghana, Kenya, South Africa, and India.

Two concrete first deals (Task-A relevant)

Telco churn intervention. An MTN / Glo / Airtel retention team runs a churn cohort through NaijaPersona Test; the engine returns predicted reaction-to-intervention-message-in-customer-register. Reduces testing of generic English scripts; lifts opt-back-in rate. **MSME credit signal.** A Nigerian neobank's underwriter pipes WhatsApp seller-buyer transcripts through NaijaUnderstand SDK; the engine reads informal Pidgin as legitimate business signal rather than noise. Expands credit reach without expanding default risk. Concrete first customer: Carbon / FairMoney / Branch.

6.1 Deployment

NaijaReviewer-8B is served in production as an OpenAI-compatible inference endpoint on **Modal**, a serverless GPU platform. The *Q4_K_M* GGUF is pulled once from HuggingFace and cached in a persistent volume, then served from a single L4 GPU through a minimal FastAPI wrapper over llama.cpp. The endpoint scales to zero when idle, so there is no standing GPU cost, and warm requests return in a few seconds. The FastAPI application reaches it through a dedicated provider abstraction, which keeps the fine-tune isolated from the frontier APIs used for ranking and persona extraction and lets any backbone be swapped per request for A/B comparison. The same engine is exposed through two end-user products, the InsideNaija synthetic-panel interface (Task A) and the ShopEasy storefront (Task B), plus an embeddable business widget, all served by one FastAPI deployment. The full build pipeline, from corpus generation through QLoRA fine-tuning, GGUF quantisation, and HuggingFace publication, is reproducible from the released Colab notebooks.

Resources. All artifacts are public.

- **Code (GitHub):** https://github.com/Mystique1337/telcoproject
- **Live app / demo:** <APP-LINK-PLACEHOLDER>
- **Model (HuggingFace):** https://huggingface.co/Shinzmann/naija-reviewer-8b-v2-Q4_K_M-GGUF
- **Training data and large artifacts (Google Drive):** <DRIVE-FOLDER-PLACEHOLDER>

7 DISCUSSION

The result-side claim of this paper — a small open-weight fine-tune that matches frontier-model performance on Nigerian-context user modelling — is interesting but expected for anyone who has worked in low-resource NLP. The non-obvious findings are methodological:

(1) The cultural prior is a measurable gap, not a soft one. Vanilla Claude with rich persona JSON already produces "Nigerian voice" on persona-aware prompts (Table 1); but on ground-truth comparison (Table 2) it recovers only 7.1 % of the cultural markers that real Nigerian reviewers used, against 13.0 % for the fine-tune. The gap is real and measurable.

(2) Rating-anchoring kills rating diversity. Our initial Stage-A heuristic predicted a rating before the LLM saw the product, locking outputs into a 3-/4-star bucket. Removing the heuristic and letting the LLM decide rating + review in a single call restored a realistic 1-5 distribution. We document this as a cautionary tale — many published LLM-as-user-simulator architectures use the same anchoring pattern and may be silently compressing rating-prediction RMSE.

(3) Detector-recall is a soft trap. Our automated register-match metric penalised the fine-tune for using Pidgin markers in Nigerian-English-declared personas. Inspection revealed the markers were authentic and human-rater-appropriate; the detector simply over-classifies into Pidgin. We report both the metric and the caveat, but other published systems that rely on automated register classifiers may be silently penalising authentic Nigerian voice.

(4) Behavioural fidelity is judge-dependent. The LLM-as-Judge pass (§5.3) shows three frontier judges disagreeing substantially on a 50-pair blind A/B set ($\kappa = 0.317$). Anyone publishing prose-authenticity claims with a single LLM judge is likely over-fitting to that judge's idiosyncrasies; the multi-judge + Fleiss-κ protocol is the minimum bar.

8 LIMITATIONS

We acknowledge five limitations. First, our held-out test split is moderate-sized: v1 was 40 product-grounded rows; v2 is 100 (filtered from 186 generated). Numbers should therefore be read as moderate-CI indications, not tight ones; an order-of-magnitude larger split is planned. We did however verify (§5.2) that the fine-tune's RMSE lead holds on both v1 and v2 and on the AgentSociety `preference_estimat` external arbiter — three independent measurements, all directionally consistent. Second, the corpus is partially synthesised from frontier models, a known risk flagged as *Lost in Simulation* [2]; we mitigate this with the AfriSenti and Iwendi-Jumia anchors but cannot fully remove the bias. Third, our 5-day build limited human evaluation to a 4-judge in-team rubric; a larger crowd-sourced study and a downstream business-impact study (e.g. telco churn lift, MSME loan default rate) is future work. The LLM-as-Judge protocol in §5.3 is a high-throughput screening surrogate, not a replacement for human ratings on the behavioural-fidelity axis. Fourth, our automated *register-match* and *cultural-marker recall* metrics are partially confounded — the review-agent prompt template explicitly tells the model which Nigerian markers to use, and the same markers appear in the GT (69 % overlap, §5.2). We therefore re-frame these numbers as "cultural-marker emission rate" rather than "authenticity", and recommend external human evaluation (`scripts/human_eval.py`) for true authenticity judgement. Fifth, the LLM-as-Judge pass shows the fine-tune *loses* on prose authenticity to frontier judges by 68 % to 32 % on majority vote. The RMSE / rating-accuracy lead is the surviving strong claim; the prose-authenticity claim is retracted.

9 CONCLUSION

The cultural prior in vanilla LLM agents is a measurable gap that compresses rating intensity, flattens register, and loses cultural markers — but a small (8 B) open-weight fine-tune, trained on 20 k Nigerian reviews and shipped alongside a structured persona representation, recovers the rating-accuracy half of that gap while remaining hostable on a single consumer GPU. The prose-authenticity half remains open: frontier judges still prefer Claude's prose by a factor of ~2 on blind A/B. We report both findings openly. The system, model, corpus, and evaluation harness are released under permissive licenses, and we invite the community to extend the cognitive persona dimensions to other underrepresented contexts. A companion paper addresses Task B (Recommendation) using the same persona representation as the input substrate.

REFERENCES

[1] David Ifeoluwa Adelani and others (Masakhane). 2021. MasakhaNER: Named Entity Recognition for African Languages. Transactions of the ACL (TACL).

[2] Anonymous. 2025. Lost in Simulation: LLM-Simulated Users are Unreliable Proxies for Human Users in Agentic Evaluations. *arXiv preprint* (2025).

[3] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, et al. 2022. Constitutional AI: Harmlessness from AI Feedback. *arXiv preprint arXiv:2212.08073* (2022).

[4] Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, et al. 2024. Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference. *arXiv preprint arXiv:2403.04132* (2024).

[5] Bing Lin et al. 2024. Orthographic Variation for Augmenting Nigerian Pidgin Training Data. *arXiv preprint arXiv:2404.18264* (2024).

[6] Shamsuddeen Hassan Muhammad, Idris Abdulmumin, Abinew Ali Ayele, et al. 2023. AfriSenti: A Twitter Sentiment Analysis Benchmark for African Languages. In *Proceedings of EMNLP*.

[7] Shamsuddeen Hassan Muhammad, David Ifeoluwa Adelani, Sebastian Ruder, et al. 2022. NaijaSenti: A Nigerian Twitter Sentiment Corpus for Multilingual Sentiment Analysis. In *Proceedings of LREC*.

[8] Kelechi Ogueji, Yuxin Zhu, and Jimmy Lin. 2021. Small Data, Big Impact: Leveraging Minimal Data for Effective Machine Translation of Low-Resource Languages (AfriBERTa). *Proceedings of the EMNLP Workshop on Multilingual Representation Learning* (2021).

[9] Wuraola Fisayo Oyewusi, Olubayo Adekanmbi, and Olalekan Akinsande. 2021. SentiLeye: A Nigerian Pidgin Sentiment Lexicon. In *Proceedings of the IJCAI AI4SG Workshop*.

[10] Lei Wang, Jingsen Zhang, Hao Yang, Zhiyuan Chen, et al. 2023. User Behavior Simulation with Large Language Model based Agents. *arXiv preprint arXiv:2306.02552* (2023).

[11] Yuxin Zhang et al. 2025. AgentCF++: User-Item Memory and Interest Groups for Recommendation Agents. *arXiv preprint arXiv:2502.13843* (2025).

[12] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, et al. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *arXiv preprint arXiv:2306.05685* (2023).